

Pranav Pandy Mohanapandian

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PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Chennai, India

Software Development Engineer | COBOL, Java, JavaScript, HTML, CSS

January 2024 – August 2024

- Developed and maintained banking transaction modules in COBOL supporting business-critical financial operations and database-driven workflows.
- Enhanced customer-facing and internal banking application screens using Java, JavaScript, HTML, and CSS.
- Resolved production defects and implemented business-rule enhancements, reducing system errors by **23%** and improving transaction throughput by **15%**.
- Performed root-cause analysis, regression testing, and incident resolution to improve stability and release quality.

VOLUNTEER EXPERIENCE

Santech America

Katy, TX

Software Development Volunteer | React Native, TypeScript, NestJS, Prisma, PostgreSQL, Azure DevOps

- Built full-stack features for **AiQMate**, an industrial field-operations platform, using React Native, NestJS, and PostgreSQL.
- Rewrote the task workflow with supervisor approve/reject/redo reviews and role-based task screens; built a Breakdown Analysis module, signature capture, and evidence uploads.
- Created a reusable mobile UI component kit for iOS and Android, and added autosave and validation to forms.
- Documented the asset health-scoring algorithm for **SanEdge**, the company's IoT edge platform that feeds its **SanHawk** predictive-maintenance cloud.

PROJECTS

Stock Trading Web & iOS App | Node.js, PostgreSQL, Swift/SwiftUI, Docker, AWS

- Engineered a full-stack stock trading platform with a shared REST backend for web and native iOS clients, integrating near real-time quotes, company profiles, and market news from **multiple financial data providers**.
- Kept portfolio holdings and cash consistent under concurrent buy/sell requests using transactional database updates.
- Containerized the application with Docker and deployed it on AWS ECS for reproducible deployments.

RoomSync – Roommate Matching & Housing Platform | Scikit-learn, REST APIs, JWT, GeoJSON

- Implemented REST APIs and JWT authentication for roommate discovery and housing search.
- Designed a recommendation engine that ranks roommate compatibility across **10+ lifestyle attributes**, with GeoJSON-based map search for listings.

Serverless YouTube Video Summarizer | AWS Lambda, Gemini API, Flask

- Architected a serverless summarization pipeline on AWS Lambda, eliminating dedicated server infrastructure and enabling on-demand execution.
- Processed transcripts from **10–60+ minute** videos and generated concise summaries through a REST endpoint integrating the YouTube Transcript and Gemini APIs.

Movie Recommender System | TensorFlow, PostgreSQL, pgvector

- Generated semantic embeddings for **35,000+ movie descriptions** and used PostgreSQL/pgvector similarity search for content-based recommendations with sub-second retrieval.

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science (GPA: 3.8/4.0)

August 2024 – May 2026

Courses: Analysis of Algorithms, Web Technologies, Machine Learning, Applied NLP, Information Retrieval

Panimalar Engineering College

Chennai, India

Bachelor of Engineering in Computer Science and Engineering (GPA: 4.0/4.0)

August 2019 – April 2023

SKILLS

Technical Skills: Data Structures & Algorithms, Object-Oriented Programming, System Design, Low-Level Design, REST API Design, Relational Database Design, Client-Server Architecture, Mobile App Development, CI/CD, Agile/Scrum

Languages & Databases: Python, Java, JavaScript, TypeScript, C/C++, SQL, Swift, COBOL, HTML/CSS, PostgreSQL, MySQL, MongoDB

Frameworks & Libraries: React Native (Expo), React, Node.js, Express.js, NestJS, Flask, SwiftUI (MVVM), Prisma ORM

Cloud & Tools: AWS (ECS, Lambda), Azure DevOps (Repos, Pipelines, Boards), Docker, Git, GitHub, Linux, Jira

AI / ML: TensorFlow, Scikit-learn, NumPy, Pandas, LangChain, Gemini API, Claude API, pgvector, Vector Embeddings, Semantic Search, Recommender Systems, Deep Learning (CNN)